Weather Forecasting Model using RNN-LSTM

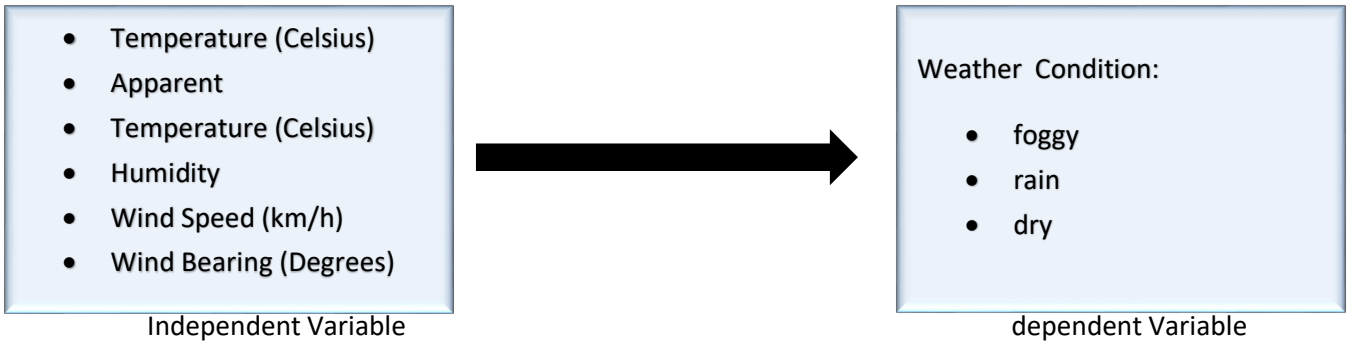
Introduction

This project aimed to anticipate a weather condition using LTSM-RNN algorithm of AI. The dataset was generated from kaggle.com.

The main Inputs are;

- Temperature (Celsius)
- Apparent Temperature(Celsius)
- Humidity
- Wind Speed (km/h)
- Wind Bearing (Degrees)

And the dependent variable was weather condition which we identify between foggy, rain, and dry



Problem Statement.

As of Dec 2020, natural disaster has affected 40,000 Malaysian population and this led to tremendous loss, the issue raises constant argument on the important weather prediction. Hence, we decided to develop a model to predict the weather. With this model, it helps to ease weather forecasting for daily life planning. On the other hand, it helps to take early necessary action to cope with unprecedented natural disaster which resulted in material losses. Besides, the model helps an Agencies like national meteorology centers to predict weather forecast.

The Ideas

Create efficient weather forecasting model with low cost and manpower using the RNN-LTSM deep learning approach. In a way, data will be train and test to ensure the compatibility of the model. The Confusion Matric was employed to monitor the accuracy of the model.

The Journey

- Data Collection
  - Weather dataset was obtained from kaggle.com from the link:
    - https://www.kaggle.com/budincsevity/szeged-weather
  - preparing data for analysis by removing or modifying data that is incorrect, incomplete, irrelevant, duplicated, or improperly formatted.
  - The data being labeled with:
    - Temperature (Celsius)
    - Apparent Temperature (Celsius)
    - Humidity
    - Wind Speed (km/h)
    - Wind Bearing (Degrees)
    - Condition
- Annotation
  - The deep learning method applied are:
    - RNN (Recurrent Neural Network):
      - A Recurrent Neural Network is a type of neural network that contains loops, allowing information to be stored within the network. RNN use due their nature in handling sequential data.
    - LSTM (Long Short-Term Memory):
      - Long short-term memory (LSTM) is an artificial recurrent neural network (RNN) has feedback connections. It can not only process single data points (such as images), but also entire sequences of data (such as speech or video).
    - Confusion Matric:
      - A confusion matrix is a summary of prediction results on a classification problem. The number of correct and incorrect predictions are summarized with count values and broken down by each class.
- Data Processing
  - Simplify raw data consisting of 11 columns into 6 column
  - Remove unnecessary info to avoid skewness and outlier's removal
  - Classify output column into 5 categories
  - Transform input-output

- Architecture and Modelling
  LSTM-RNN consist of:
  1. Seed : 123
  2. Hidden Layer: 2
  3. Regularization: L2
  4. Learning Rate: 0.001
  5. Activation Function: Tanh (Hidden Layer), Softmax (Output Layer)

Possible Improvement

- Train model using Malaysian Weather Data
- Use more datasets to improve generality
- Tried with more advanced model and architecture
    - Different number of hidden layers
    - Different number learning rates
    - Number of neurons